WIRESHARK

Ethical Hacking Study Notes

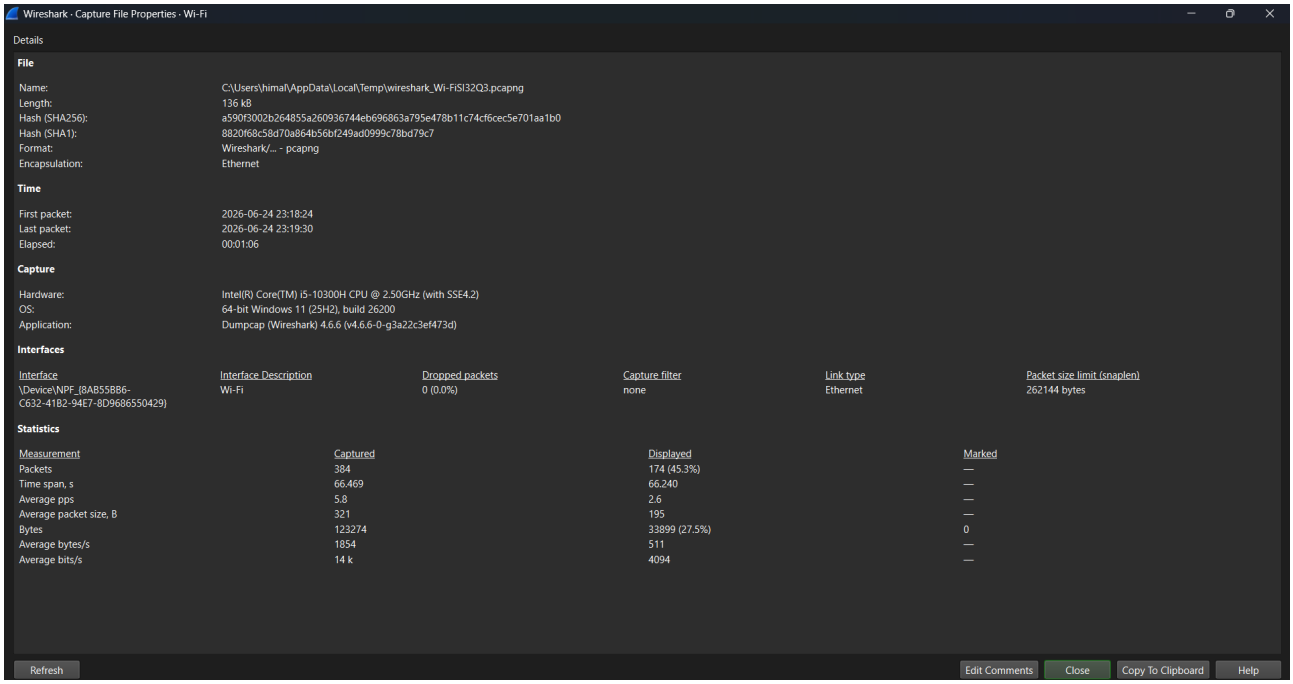
Complete Session Reference — Packet Analysis & Network Forensics

Topics Covered	15 Sections
Wireshark Screens	Protocol Hierarchy, Conversations, Endpoints, DNS, HTTP, TCP Graphs...
Key Skills	Packet filtering, traffic analysis, protocol identification
Level	Beginner to Intermediate Ethical Hacking

TABLE OF CONTENTS

#	Section	Topic
1	Capture File Properties	File metadata, SHA256 hash verification
2	Resolved Addresses	MAC addresses & OUI lookup
3	Protocol Hierarchy	Traffic breakdown by protocol
4	Conversations	Who talked to who
5	Endpoints	Every IP in the capture
6	DNS Packets	Domain name analysis
7	Packet Lengths	Size distribution analysis
8	I/O Graphs	Traffic over time
9	Flow Graph	Visual packet sequence diagram
10	TCP & TLS Handshakes	Connection establishment
11	HTTP Analysis	HTTP packet counter & requests
12	TCP Sequence Graphs	Stevens & tcptrace graphs
13	Round Trip Time	Latency analysis
14	Key Filters	Essential Wireshark filters
15	Analyst Mindset	How to think like a real analyst

SECTION 1 — CAPTURE FILE PROPERTIES



■ What is it?

Accessed via Statistics → Capture File Properties. This is the metadata/summary page of your entire capture file — overview before diving into actual packets.

■ Key Sections

Section	What it shows	Why it matters
File	Name, size, SHA256/SHA1 hash, format	Verify file integrity
Time	First packet, last packet, elapsed time	Confirm capture duration
Capture	Hardware, OS, application used	Know the environment
Interfaces	Which network interface was used	Wi-Fi vs Ethernet matters
Statistics	Packets captured vs displayed, bytes, speed	Overview of traffic volume

■ SHA256 Hash — File Integrity Verification

The SHA256 hash is like a fingerprint of the file. If even one byte changes, the entire hash changes completely. Wireshark calculates and displays it — it does NOT compare automatically. You must compare manually.

Scenario	Action
Someone gives you a pcap file	Compare their stated hash vs what Wireshark shows
You download from internet	Compare website's published hash vs your file's hash
Hashes match	File is untampered ■
Hashes differ	File was modified or corrupted ■

■ Command Line Verification

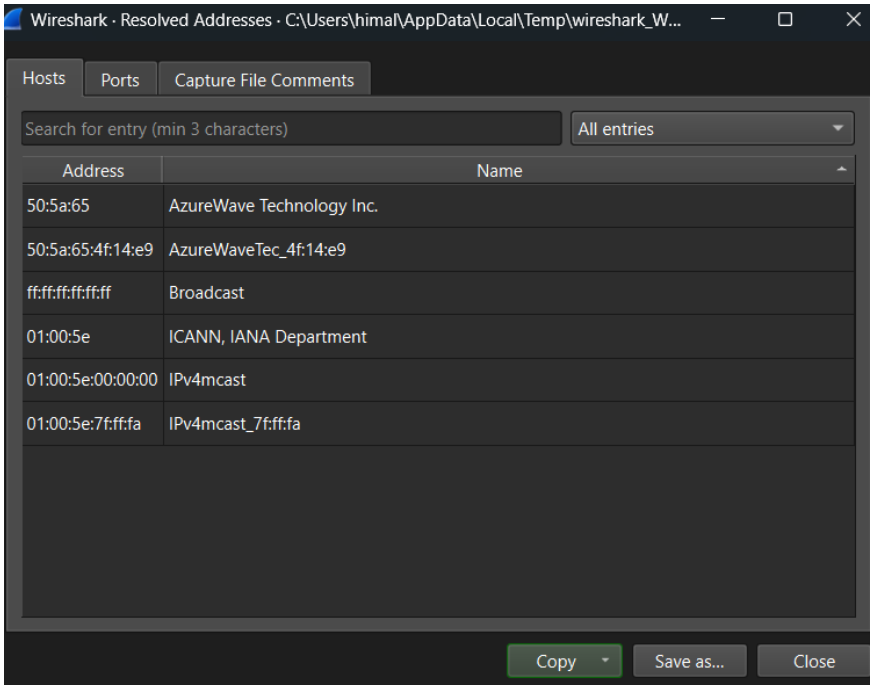
```
Windows: certutil -hashfile capture.pcapng SHA256
```

```
Linux: sha256sum capture.pcapng
```

■ Always verify hash before trusting a pcap file from someone else or downloaded from the internet!

■ 0.0% dropped packets = clean, complete capture. High drop rate = missing evidence.

SECTION 2 — RESOLVED ADDRESSES



■ What is it?

Accessed via Statistics → Resolved Addresses. Shows MAC addresses translated into human-readable manufacturer names using the OUI (Organizationally Unique Identifier) database.

■ MAC vs IP Address

Property	MAC Address	IP Address
Looks like	50:5a:65:4f:14:e9	192.168.1.5
Separator	Colons (:)	Dots (.)
Assigned by	Manufacturer (permanent)	Router (dynamic/changes)
Works at	Layer 2 (local network only)	Layer 3 (across internet)
Think of it as	Permanent ID card	Home address (can change)

■ Special MAC Addresses

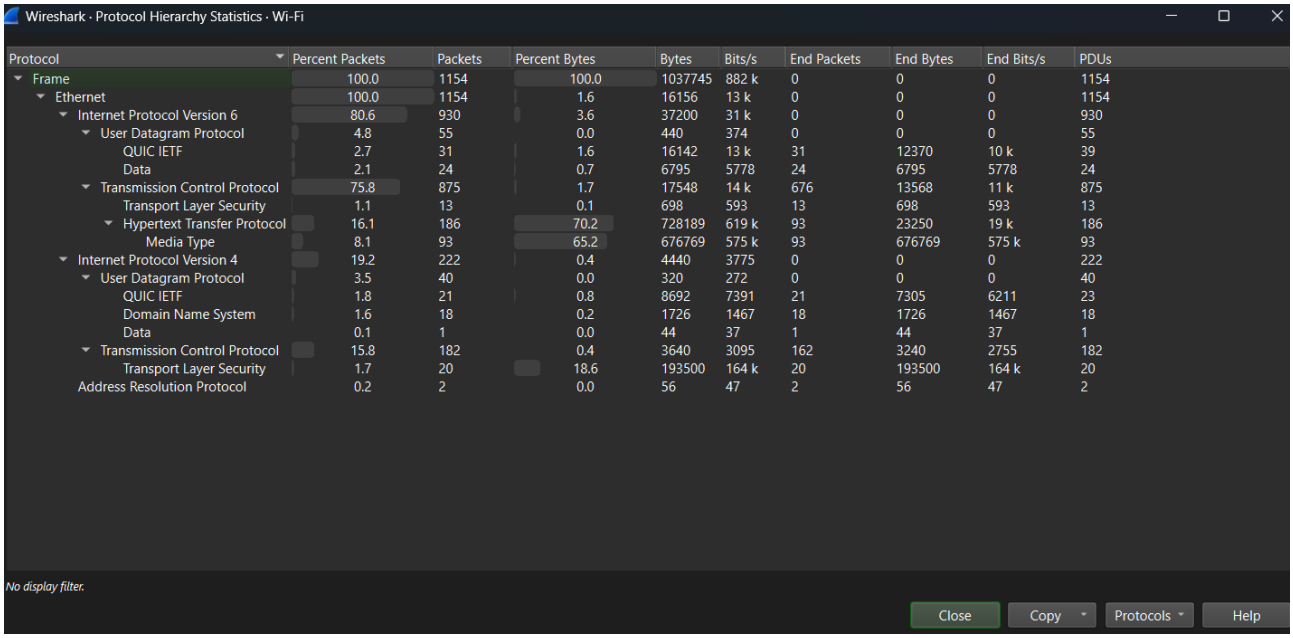
MAC Address	Name	Meaning
ff:ff:ff:ff:ff:ff	Broadcast	Packet sent to ALL devices on network
01:00:5e:xx:xx:xx	IPv4 Multicast	Sent to a group of subscribed devices
50:5a:65:xx:xx:xx	AzureWave	Wi-Fi chip (common in laptops)

■ Ethical Hacking Use

- Identify device manufacturer from MAC alone during LAN recon
- Detect MAC spoofing — if MAC says Intel but device claims to be a printer, suspicious!
- Most useful during Wi-Fi/LAN penetration testing

■ This screen only shows MAC addresses — NOT IP addresses. IPs are in Conversations and Endpoints.

SECTION 3 — PROTOCOL HIERARCHY



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	1154	100.0	1037745	882 k	0	0	0	1154
Ethernet	100.0	1154	1.6	16156	13 k	0	0	0	1154
Internet Protocol Version 6	80.6	930	3.6	37200	31 k	0	0	0	930
User Datagram Protocol	4.8	55	0.0	440	374	0	0	0	55
QUIC IETF	2.7	31	1.6	16142	13 k	31	12370	10 k	39
Data	2.1	24	0.7	6795	5778	24	6795	5778	24
Transmission Control Protocol	75.8	875	1.7	17548	14 k	676	13568	11 k	875
Transport Layer Security	1.1	13	0.1	698	593	13	698	593	13
Hypertext Transfer Protocol	16.1	186	70.2	728189	619 k	93	23250	19 k	186
Media Type	8.1	93	65.2	676769	575 k	93	676769	575 k	93
Internet Protocol Version 4	19.2	222	0.4	4440	3775	0	0	0	222
User Datagram Protocol	3.5	40	0.0	320	272	0	0	0	40
QUIC IETF	1.8	21	0.8	8692	7391	21	7305	6211	23
Domain Name System	1.6	18	0.2	1726	1467	18	1726	1467	18
Data	0.1	1	0.0	44	37	1	44	37	1
Transmission Control Protocol	15.8	182	0.4	3640	3095	162	3240	2755	182
Transport Layer Security	1.7	20	18.6	193500	164 k	20	193500	164 k	20
Address Resolution Protocol	0.2	2	0.0	56	47	2	56	47	2

■ What is it?

Accessed via Statistics → Protocol Hierarchy. Shows every protocol in your capture as a tree, with percentage of traffic per protocol. One of the most important screens for ethical hacking.

■ Tree Structure

```
Frame (everything 100%)
Ethernet
IPv6 (80.6%) → UDP → QUIC / TCP → TLS / HTTP
IPv4 (19.2%) → UDP → DNS / QUIC / TCP → TLS
```

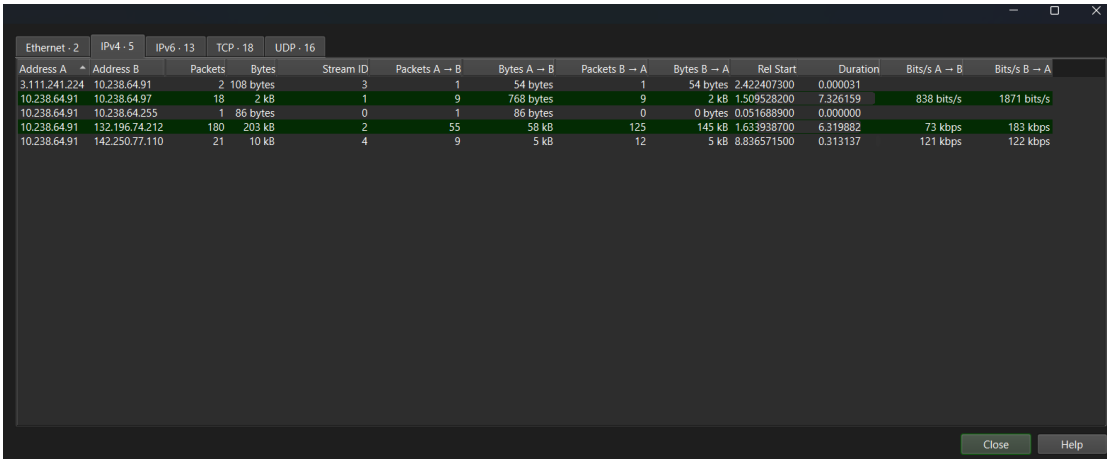
■ Key Protocols

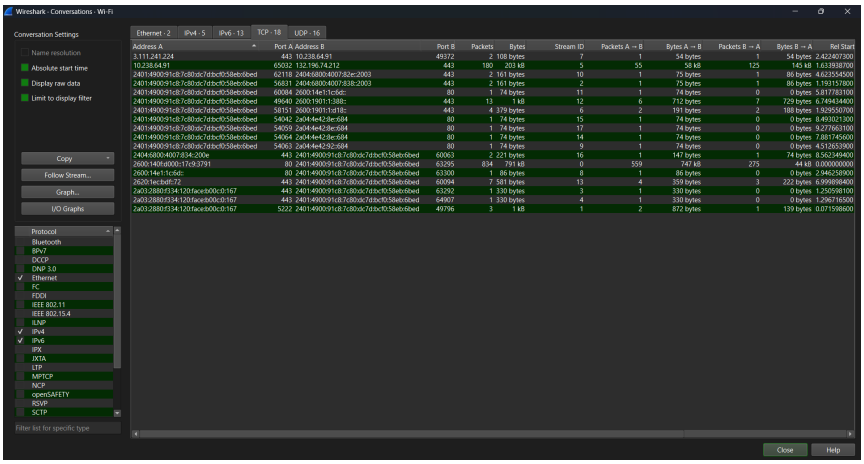
Protocol	Port	Encrypted?	Hacking Significance
HTTP	80	NO ■	Readable — usernames, passwords, data visible
HTTPS/TLS	443	YES ■	Encrypted — see IPs but not content
DNS	53	Usually NO	Reveals every domain name visited
QUIC	443 UDP	YES ■	Google/YouTube encrypted UDP
ARP	N/A	NO ■	Network discovery — who has what IP

■ HTTP is the golden ticket — if present, filter 'http' in main window to read content!

■ High DNS percentage = map every site a target visits, even with encrypted HTTPS traffic.

SECTION 4 — CONVERSATIONS





■ What is it?

Accessed via Statistics → Conversations. Shows every pair of devices that communicated. Available across Ethernet, IPv4, IPv6, TCP, and UDP layers.

■ The 5 Tabs

Tab	Shows	Key Info
Ethernet	MAC to MAC	Device-level communication
IPv4	IPv4 to IPv4	Actual IP addresses
IPv6	IPv6 to IPv6	Modern IPv6 addresses
TCP ■	TCP connections with ports	Ports reveal service type
UDP	UDP connections with ports	DNS (53), QUIC (443)

■ Key Port Numbers

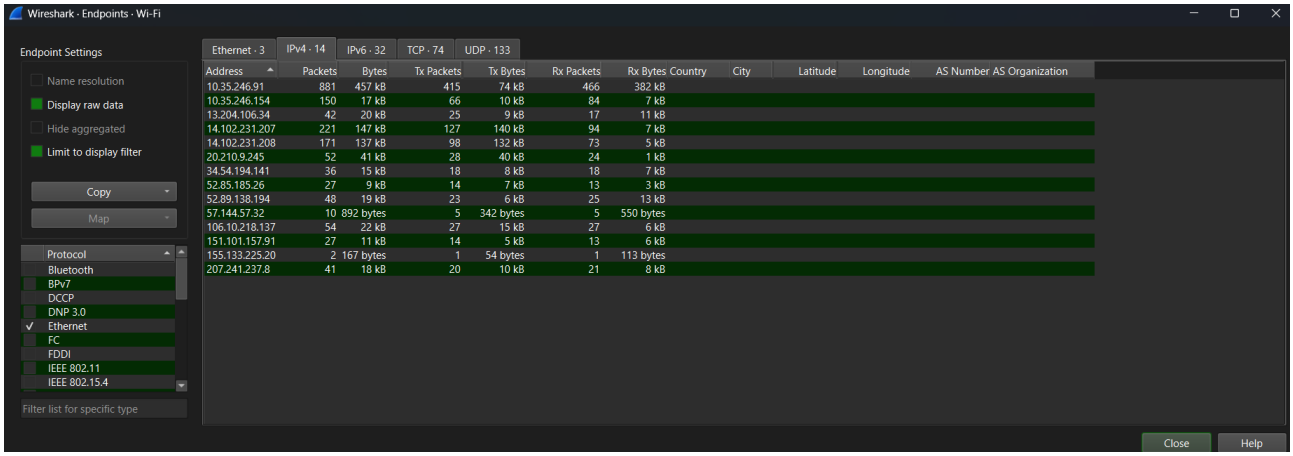
Port	Protocol	Meaning
80	HTTP	Unencrypted web — readable!
443	HTTPS/TLS	Encrypted web traffic
53	DNS	Domain name lookups

22	SSH	Secure remote access
21	FTP	File transfer — unencrypted!
5222	XMPP	Chat protocol

■ Ethical Hacking Tips

- Sort by Bytes to find heaviest transfers — possible data exfiltration
- Look for port 80 conversations — unencrypted readable content
- Right-click any conversation → Follow Stream — read the entire exchange!

SECTION 5 — ENDPOINTS



■ What is it?

Accessed via Statistics → Endpoints. Lists every single IP/MAC that appeared in your capture, individually, with traffic stats. Complete roster of every device involved.

■ Conversations vs Endpoints

Feature	Conversations	Endpoints
What it shows	A ↔ B pairs (who talked to who)	Each IP/MAC individually
Analogy	Phone call log (both parties)	Contacts list (one entry each)
Use case	Analyze specific connections	Map all devices in capture

■ GeoIP — Location Mapping

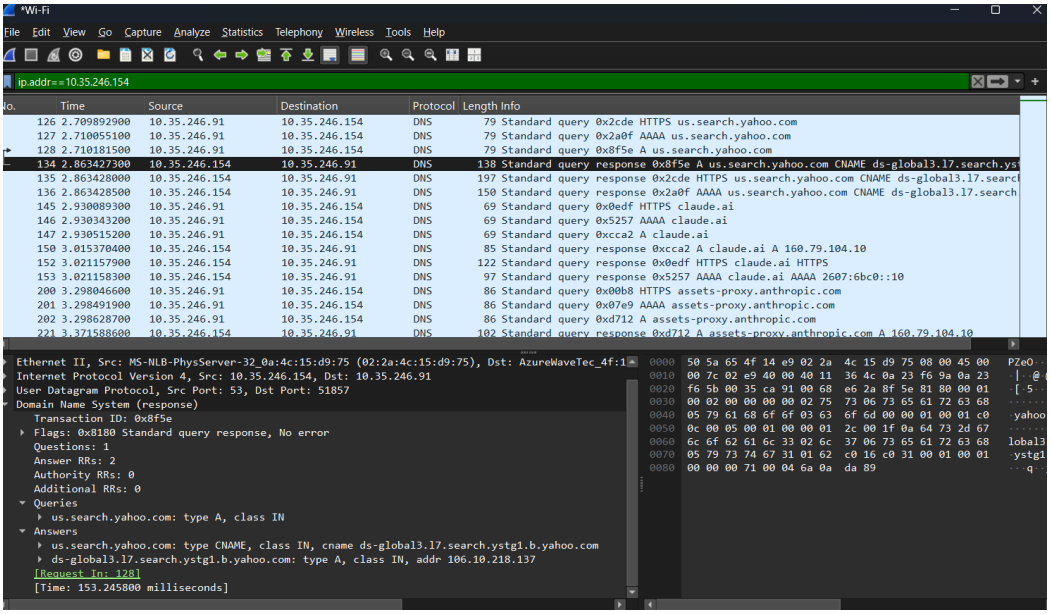
Country, City, Latitude, and Longitude columns are empty by default. Enable via Edit → Preferences → Name Resolution → MaxMind database. Once set up, the Map button plots all IPs on a world map.

■ GeoIP is extremely useful for spotting traffic going to unexpected countries — potential data exfiltration!

■ Ethical Hacking Use

Scenario	How Endpoints Helps
Recon on a target	See every server their device contacts
Malware analysis	Spot unknown IPs a device is calling home to
Network mapping	Build complete picture of all devices
Find your own IP	Usually the IP with highest packet count

SECTION 6 — DNS PACKET ANALYSIS

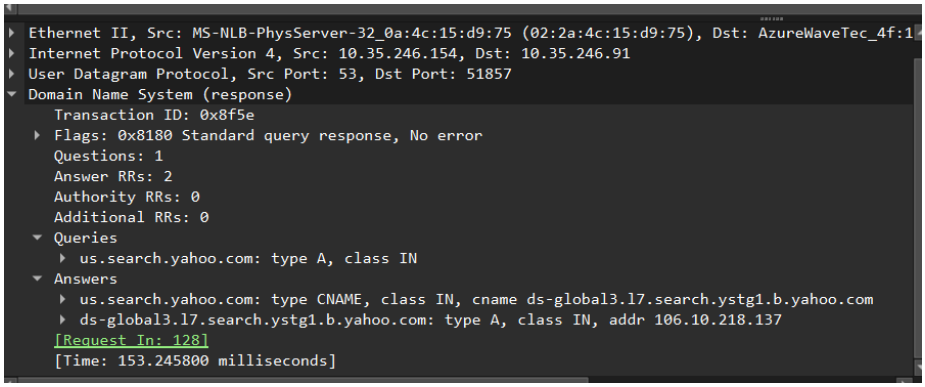


■ What is DNS?

DNS (Domain Name System) is the phonebook of the internet. When you type 'claude.ai', your device asks a DNS server for the IP address. DNS traffic is extremely valuable in ethical hacking.

■ Even if ALL traffic is encrypted with HTTPS/TLS, DNS queries usually reveal domain names in plaintext. DNS = the truth about browsing activity.

■ Anatomy of a DNS Packet



Field	Value in Example	Meaning
Transaction ID	0x8f5e	Unique ID matching request to response
Flags	0x8180 Standard query response	This is a RESPONSE with no error
Questions	1	One domain was asked about
Answer RRs	2	Two answers returned
Query	us.search.yahoo.com type A	What your device asked
CNAME Answer	us.search.yahoo.com → ds-global3.l7...	Alias/redirect to another name
A Record	ds-global3... → 106.10.218.137	Final IP address
Time	153.245800 ms	How long DNS took to respond

■ DNS Record Types

Type	Meaning	Example
A	IPv4 address	claude.ai → 160.79.104.10
AAAA	IPv6 address	claude.ai → 2607:6bc0::10
CNAME	Alias/redirect	www.example.com → example.com
MX	Mail server	gmail.com → mail server IP
SOA	Authority record	Who manages this domain

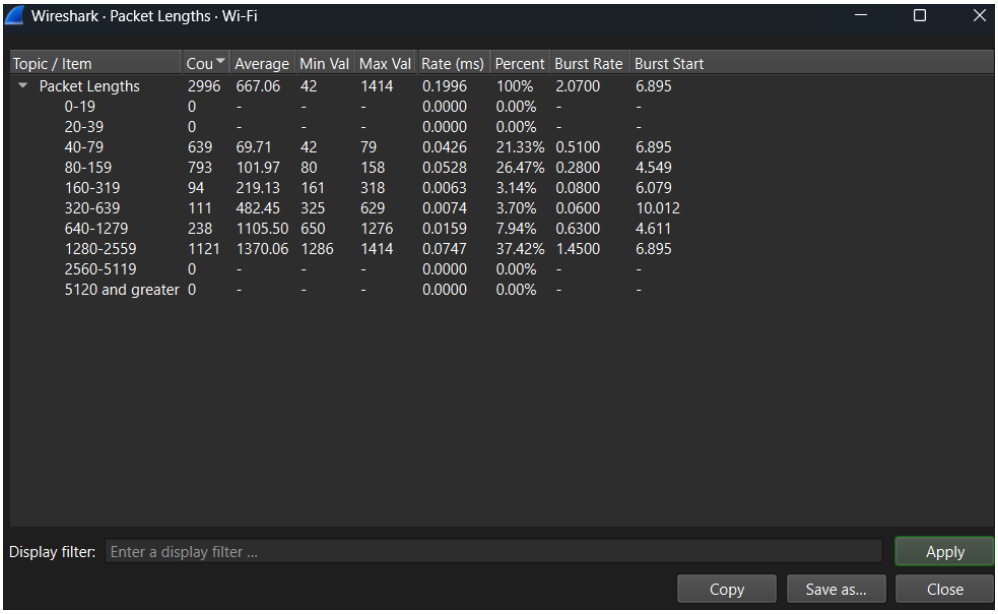
■ Where to See Domain Names

Location	What you can see
Telephony → DNS Statistics	Counts, percentages, response times ONLY — NO domain names
Main packet list + 'dns' filter	Actual domain names, full query details ■

Why you see unexpected domains: websites secretly load resources from many third-party servers — ads, analytics, CDNs, fonts, tracking scripts.

■ Filter: 'dns' shows all queries. 'dns.qry.name contains google' finds specific domains.

SECTION 7 — PACKET LENGTHS



Topic / Item	Cou	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
Packet Lengths	2996	667.06	42	1414	0.1996	100%	2.0700	6.895
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	639	69.71	42	79	0.0426	21.33%	0.5100	6.895
80-159	793	101.97	80	158	0.0528	26.47%	0.2800	4.549
160-319	94	219.13	161	318	0.0063	3.14%	0.0800	6.079
320-639	111	482.45	325	629	0.0074	3.70%	0.0600	10.012
640-1279	238	1105.50	650	1276	0.0159	7.94%	0.6300	4.611
1280-2559	1121	1370.06	1286	1414	0.0747	37.42%	1.4500	6.895
2560-5119	0	-	-	-	0.0000	0.00%	-	-
5120 and greater	0	-	-	-	0.0000	0.00%	-	-

■ What is it?

Accessed via Statistics → Packet Lengths. Shows size distribution of all packets in your capture, broken into size ranges.

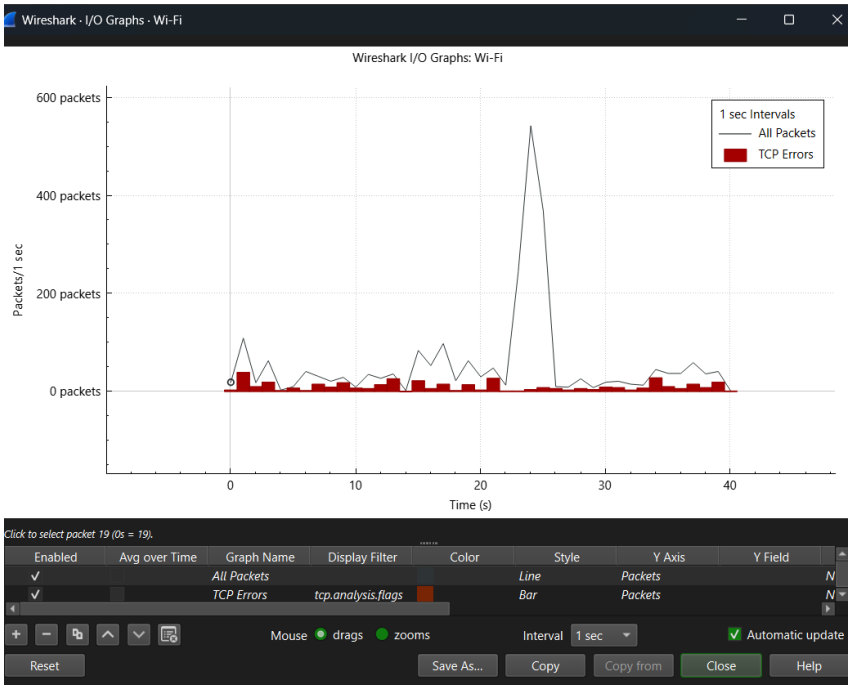
■ Size Ranges and Meanings

Size Range	Typical Contents	Significance
40-79 bytes	ACK packets, TCP handshake flags	Control traffic
80-159 bytes	DNS queries, small HTTP requests	Normal control traffic
320-639 bytes	Medium data chunks	Normal
640-1279 bytes	Larger responses	Normal data transfer
1280-2559 bytes	Full data packets (MTU-sized)	Main data transfer
2560+ bytes	Jumbo frames	Unusual on most networks

■ Ethical Hacking Patterns

Pattern	Possible Interpretation
Lots of tiny packets (40-79 bytes)	Port scanning / reconnaissance ■
Lots of huge packets (1280+ bytes)	File transfer or data exfiltration ■
Mixed sizes	Normal web browsing ■
All same size packets	Possible automated/scripted traffic

SECTION 8 — I/O GRAPHS



■ What is it?

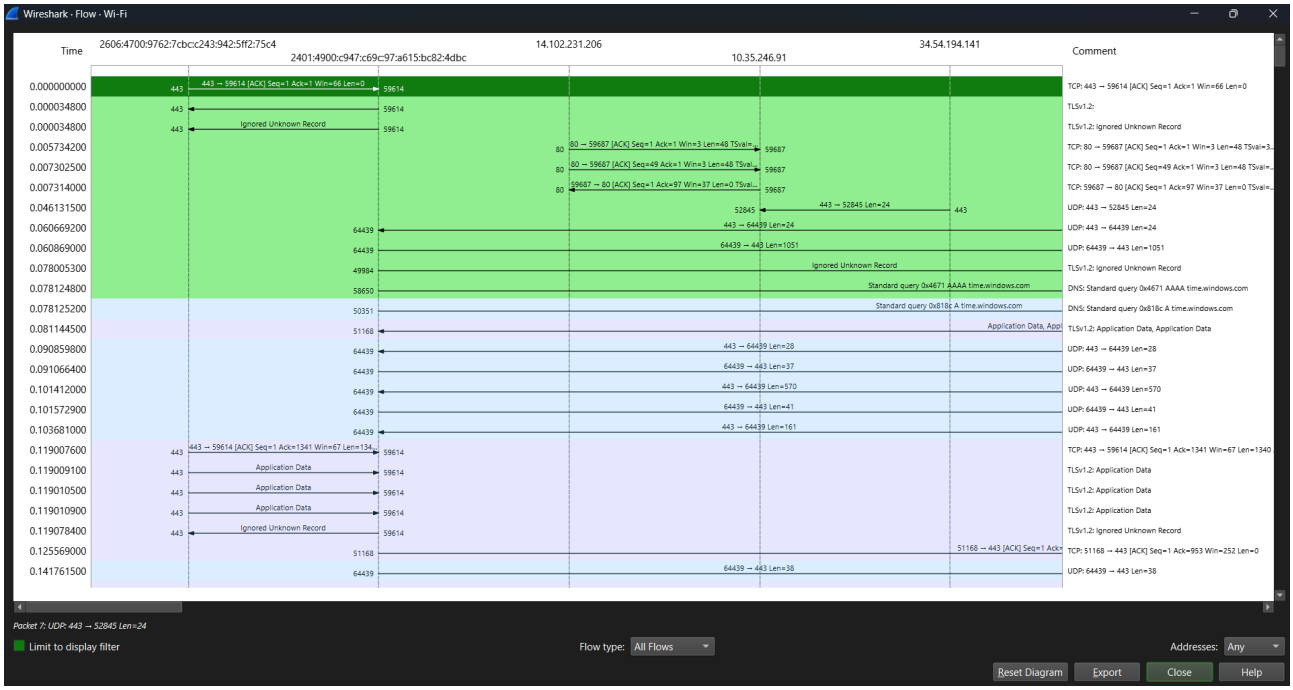
Accessed via Statistics → I/O Graphs. Shows network traffic volume visually over time. X axis = time, Y axis = packets per second.

■ Traffic Patterns

Pattern	What it Suggests
Large sustained spike	DDoS attack in progress ■
Many tiny rapid spikes	Port scanning / reconnaissance ■
Sudden large outgoing burst	Possible data exfiltration ■
Regular periodic spikes (e.g. every 60s)	Malware calling home (C2 beacon) ■
Irregular random spikes	Normal web browsing ■
Consistent red TCP error bars	Network problems / packet loss

■ The spike at ~24 seconds in your capture reaching 550 packets/sec was likely YouTube buffering.

SECTION 9 — FLOW GRAPH



■ What is it?

Accessed via Analyze → Flow Graph. Shows every packet as an arrow between devices in chronological order. Each column = one device. Each arrow = one packet. Arrow direction = which way packet traveled.

■ Identifying Client vs Server

Indicator	Client	Server
IP type	Private (10.x, 192.168.x)	Public IP
Port number	Random high port (49000+)	Well-known port (80, 443)
Who initiates	Sends first SYN	Responds with SYN-ACK
Your capture	10.35.246.91:56665	14.102.231.204:80

■ Ethical Hacking Use

Use Case	What to Look For
Verify TCP handshake	SYN → SYN-ACK → ACK sequence
Verify TLS success	Application Data flowing after handshake
Detect attacks	RST packets, repeated SYNs
Trace malware	Connections to unknown IPs, odd protocols
Incident response	Exact timeline of all events

SECTION 10 — TCP & TLS HANDSHAKES

■ TCP 3-Way Handshake

```
Step 1: Client → SYN → Server (I want to connect)
Step 2: Client ← SYN-ACK ← Server (OK, I am ready)
Step 3: Client → ACK → Server (Great, let us go!)
```

■ TLS Handshake (happens after TCP)

```
→ Client Hello (Here are the encryption methods I support)
← Server Hello (Let us use THIS encryption method)
← Certificate (Here is my identity proof)
→ Key Exchange (Let us agree on a shared secret key)
Both Application Data (Encrypted data flowing = SUCCESS!)
```

■ Success vs Failure

What you see	Meaning
SYN → SYN-ACK → ACK all present	TCP handshake successful ■
Application Data flowing	TLS handshake successful ■
Only SYN, no SYN-ACK	Connection refused or target unreachable ■
TLS Alert after Certificate	TLS negotiation failed ■
RST packet appears	Connection forcefully reset ■
Repeated SYNs to same port	Possible port scan ■

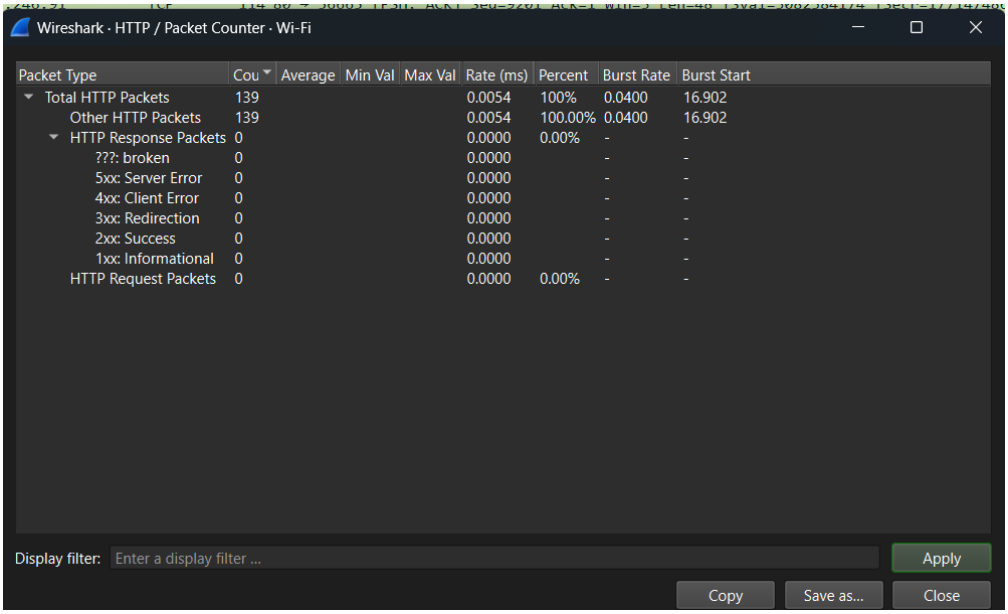
■ Two Types of ACK — Do Not Confuse Them!

■ There are TWO different ACKs. Beginners often confuse them!

ACK Type	Direction	When	Purpose
Handshake ACK	Client → Server	Once, during setup	Completes 3-way handshake
Data ACK	Both directions constantly	During data transfer	Confirms data received

■ Always start Wireshark BEFORE doing any activity so you capture the complete handshake from the beginning.

SECTION 11 — HTTP ANALYSIS



■ HTTP Packet Counter (Telephony → HTTP → Packet Counter)

Shows breakdown of HTTP packets: requests (GET, POST), responses (200 OK, 404 Not Found, 500 Server Error), and other HTTP traffic.

■ HTTP Response Codes

Code	Category	Hacking Significance
1xx	Informational	Usually benign
2xx	Success — 200 OK	Normal successful requests
3xx	Redirection — 301, 302	Following redirects
4xx	Client Error — 404, 403	Recon — finding what exists
5xx	Server Error — 500	Server struggling — exploit potential

■ Why HTTP Screens May Show Zero

If you only visited HTTPS sites the HTTP screens show zeros because:

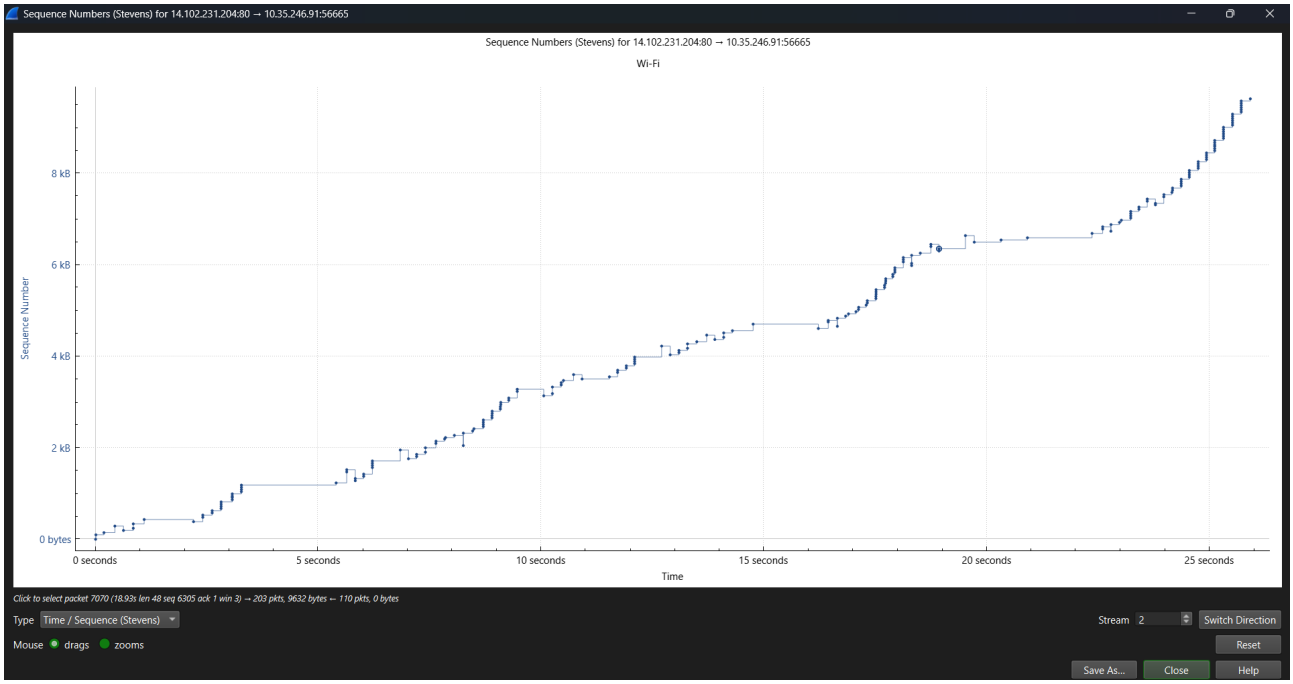
- All modern sites force HTTPS — HTTP redirected to HTTPS
- HTTPS runs inside TLS — Wireshark cannot read the HTTP inside
- HTTP/2 and HTTP/3 run entirely within TLS on port 443

■ When HTTP Screens ARE Useful

Screen	Useful When
Packet Counter	Counting error codes on a target server
HTTP Requests	Seeing exactly what URLs were requested
Load Distribution	Which servers handle most HTTP traffic
Request Sequences	Spotting patterns in how a target browses

■ HTTP analysis only works for unencrypted HTTP. HTTPS/TLS traffic cannot be read without the private key.

SECTION 12 — TCP SEQUENCE NUMBER GRAPHS



■ Stevens Graph — Time/Sequence (right-click packet → Statistics)

X axis = time, Y axis = sequence number (bytes transferred). Shows how data transfer progressed over time.

■ Reading the Graph

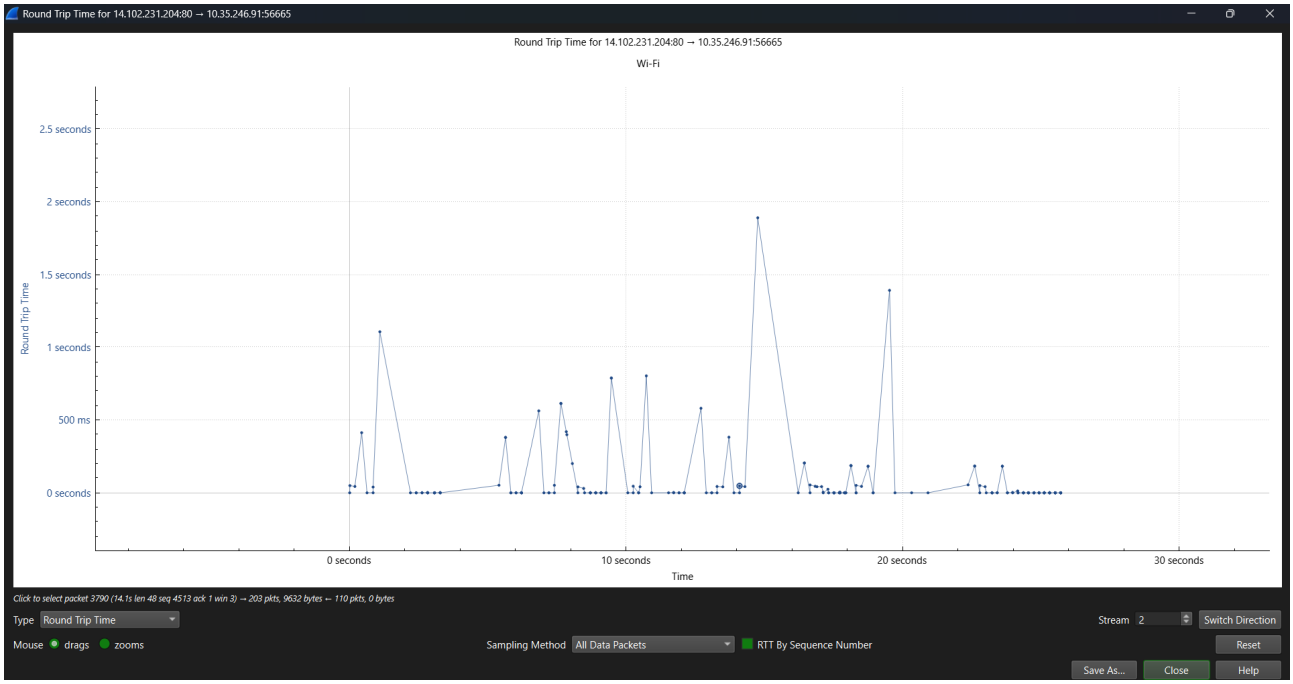
What you see	What it means	Next step
Smooth diagonal line	Healthy continuous transfer ■	No action needed
Staircase pattern	Bursty delivery — data in chunks	Normal for web browsing
Long flat sections	Transfer stalled	Check RTT and retransmissions
Backwards jumps	Retransmissions — packet resent	Investigate packet loss
Completely flat	Connection stalled entirely	Check for zero window or RST

■ All 5 TCP Graph Types

Graph	What it Shows	Best For
Time Sequence (Stevens)	Sequence numbers over time	Overall transfer health
Time Sequence (tcptrace)	More detailed — shows ACKs too	Spotting retransmissions
Throughput	Data speed (bytes/sec) over time	Speed drops, bottlenecks
Round Trip Time	Latency per packet over time	Latency spikes, congestion
Window Scaling	TCP receive window size	Buffer/flow control issues

■ A graph alone is a CLUE GENERATOR, not a final answer. Flat section does NOT automatically mean packet loss — investigate further!

SECTION 13 — ROUND TRIP TIME (RTT)



■ What is it?

Shows how long each packet takes to get acknowledged. Low RTT = fast network. High RTT = slow/congested/problematic. Spikes reveal periods of significant delay.

■ RTT Benchmarks

RTT Value	Rating	Meaning
1-20ms	Excellent ■	Very fast — nearby server
20-100ms	Good ■	Normal internet traffic
100-300ms	Acceptable	Some distance or load
300-500ms	Slow ■	Network congestion
500ms-1s	Very slow ■	Significant issues
1s+	Critical ■	Serious problems or retransmission

■ RTT Spikes — Possible Causes

■ RTT spikes do NOT automatically mean packet loss! Multiple causes exist:

Cause	Description
Retransmission	Packet lost and resent — wait time shows as spike
Network congestion	Too much traffic on the route
Wi-Fi interference	Signal quality dropped temporarily
Device busy	Receiver processing something else
Delayed ACKs	Receiver intentionally delays ACK (TCP optimization)
Power saving	Device CPU throttled momentarily

■ Building a Strong Conclusion

```
RTT spikes
+ Flat sections in Stevens graph
+ Retransmission packets visible in packet list
= Strong evidence of packet loss or significant network delay
```

■ Analyst Vocabulary

Overconfident ■	Careful & Defensible ■
'Packet loss happened'	'Suggests packet loss'
'Proves there was an attack'	'Evidence consistent with...'
'Definitely caused by X'	'Correlates with X'
'This IS malware'	'This warrants further investigation'

SECTION 14 — ESSENTIAL WIRESHARK FILTERS

■ Protocol Filters

Filter	What it Shows
dns	All DNS traffic — see domain names
http	Plain unencrypted HTTP traffic
tls	All TLS encrypted traffic
tls.handshake	Only TLS handshake packets
tcp	All TCP traffic
udp	All UDP traffic
arp	ARP requests and replies
quic	QUIC protocol (Google/YouTube)

■ IP & Port Filters

Filter	What it Shows
ip.addr == 10.35.246.91	All traffic to/from specific IP
ip.src == 10.35.246.91	Only traffic FROM that IP
ip.dst == 10.35.246.91	Only traffic TO that IP
tcp.port == 80	All TCP traffic on port 80
tcp.port == 443	All HTTPS traffic
udp.port == 53	All DNS traffic
tcp.port == 80 && !http	Port 80 traffic that is NOT HTTP

■ TCP Flag Filters

Filter	What it Shows
tcp.flags.syn == 1	All SYN packets (connection attempts)
tcp.flags.rst == 1	RST packets — forceful connection resets ■
tcp.flags.fin == 1	FIN packets — graceful connection closes
tcp.analysis.retransmission	All retransmitted packets
tcp.analysis.duplicate_ack	Duplicate ACKs — sign of packet loss

■ HTTP & DNS Filters

Filter	What it Shows
http.request.method == GET	HTTP GET requests
http.request.method == POST	HTTP POST requests (may contain passwords!)
http.request.method == CONNECT	CONNECT tunnel requests

http.response.code == 404	Not Found responses
dns.qry.name contains google	DNS queries containing 'google'
dns.qry.name == claude.ai	DNS queries for specific domain

■ Combining Filters

```
&& = AND (both conditions must be true)
|| = OR (either condition can be true)
! = NOT (exclude this condition)
Example: ip.addr == 10.35.246.91 && tcp.port == 443
Example: http || dns
Example: tcp && !tls (TCP that is not TLS encrypted)
```

SECTION 15 — THE ANALYST MINDSET

■ Every graph, filter, and statistic is a CLUE GENERATOR — not a final answer. Build conclusions from multiple corroborating pieces of evidence.

■ Wrong vs Right Approach

Wrong Approach ■	Right Approach ■
See one thing → jump to conclusion	Gather multiple pieces of evidence first
One filter = the answer	Multiple filters = verify and cross-check
Trust your first interpretation	Question it, test it, verify it
Overconfident language	Use: 'suggests', 'correlates', 'evidence of'
Ignore alternative explanations	Consider all possible causes
Graph = final answer	Graph = clue, tells you where to look next

■ The Investigation Chain

```
Step 1: Protocol Hierarchy → What protocols exist?
Step 2: Conversations → Who is talking to who?
Step 3: Endpoints → What IPs are involved?
Step 4: DNS filter → What domains were visited?
Step 5: Follow TCP Stream → What was actually said?
Step 6: TCP Graphs → Is the connection healthy?
Step 7: Form conclusion with careful, defensible language
```

■ Priority Reference

Priority	Focus Area	Why
Must ■	IPs, protocols, packet content	Core of all analysis
Must ■	Follow TCP/UDP Stream	Read actual conversations
Important ■	Conversations & Endpoints	Map the network
Important ■	DNS queries	See all domains visited
Situational ■	TCP graphs (Stevens, RTT)	Connection health clues
Situational ■	MAC addresses	Device ID on LAN
Background ■	Capture File Properties	File metadata/integrity

■ Quick Reference

Question	Where to Look
What protocols are in use?	Protocol Hierarchy
Who is talking to who?	Conversations
What IPs are involved?	Endpoints
What websites were visited?	DNS filter in main window

Is traffic encrypted?	Protocol Hierarchy — TLS vs HTTP
What was actually sent?	Follow TCP Stream
Is the connection healthy?	TCP Sequence + RTT graphs
Is this file tampered?	Capture File Properties → SHA256

Keep learning, keep questioning, keep verifying! ■